Definitions, Theorem 6, and Lemma 7
Nonzero plant initial conditions

The following replacement text follows the notation, numbering, and proof structure of the supplied ACC paper [1]. Throughout, G maps an input to the output for the specified initial state $x(0)$; this initial state is arbitrary and is never set to zero. Thus an initialized linear plant generally defines an affine input–output operator. The multiplier Ψ has its standard zero initialization, and Δ has the initialization specified by its input–output operator. No conclusion about arbitrary hidden states of Δ is implied.

Notation and signal spaces

For a Hilbert space $\mathcal{H}$, let $L_{\mathcal{H}}$ denote the space of square-integrable functions $u : [0, \infty) \to \mathcal{H}$, with

$$\langle u, v \rangle_{L_{\mathcal{H}}} = \int_0^\infty \langle u(t), v(t) \rangle_{\mathcal{H}} \, dt.$$

Let $L_{e,\mathcal{H}}$ denote the extended space of functions for which $\int_0^T \|u(t)\|_{\mathcal{H}}^2 \, dt$ is finite for every $T \geq 0$. Given $a < b$ and $n = (n_1, n_2)$, write

$$Z^n := \mathbb{R}^{n_1} \times L_2^{n_2}[a, b],$$
$$L^n_{e,[a,b]} := L_{e,Z^n}, \qquad L^n_Z := L_{Z^n}.$$

The norm on Z^n is induced by the Euclidean inner product and the spatial L_2 inner product. The stacked signal $\begin{bmatrix} u \\ v \end{bmatrix}$ denotes componentwise concatenation in Z^{n+m}. The subscript L denotes the appropriate time-signal Hilbert norm, and Z the appropriate state or spatial Hilbert norm. Define the time-truncation operator by

$$(P_\tau y)(t) = \begin{cases} y(t), & 0 \leq t \leq \tau, \\ 0, & t > \tau. \end{cases}$$

The bounded PI operators Π_4 and their adjoints are as in the paper. The storage operator $\mathcal{P}$ is distinct from the truncation P_τ.

Definition 1 (Operators)**.** Let $\mathcal{H}, \mathcal{G}$ be Hilbert spaces. For each fixed admissible initial condition $x(0)$, an operator $G : L_{e,\mathcal{H}} \to L_{e,\mathcal{G}}$ is

1. *Causal* if $P_\tau G = P_\tau G P_\tau$ for any $\tau \geq 0$. The same initial condition is used on both sides.
2. *Bounded on $L_{\mathcal{H}}$ allowing for initial conditions* if there exist $C_G, c_G \geq 0$ such that, for all $v \in L_{\mathcal{H}}$ and all admissible $x(0)$,
$$\|Gv\|_{L_{\mathcal{G}}} \leq C_G \|v\|_{L_{\mathcal{H}}} + c_G \|x(0)\|_Z .$$
3. *Bounded on $L_{e,\mathcal{H}}$ allowing for initial conditions* if there exist $C_G, c_G \geq 0$ such that, for all $v \in L_{e,\mathcal{H}}$, all admissible $x(0)$, and all $\tau \geq 0$,
$$\|P_\tau G v\|_{L_{\mathcal{G}}} \leq C_G \|P_\tau v\|_{L_{\mathcal{H}}} + c_G \|x(0)\|_Z .$$
4. *Incrementally $L_{e,\mathcal{H}}$-bounded* if there exists $C \geq 0$ such that, for all $v, u \in L_{e,\mathcal{H}}$,
$$\|P_\tau(Gv - Gu)\|_{L_{\mathcal{G}}} \leq C \|P_\tau(v - u)\|_{L_{\mathcal{H}}}, \quad \tau \geq 0.$$
Here Gv and Gu have the same initial condition; the bound is uniform over the admissible initial conditions under consideration.

The constants in items 2 and 3 are independent of $x(0)$.

Items 2 and 3 are the modifications to the paper's boundedness definitions. For an operator without an initial-state contribution, take $c_G = 0$ to recover the original bounds. In particular, this convention is used for Ψ. The notation $\mathcal{L}(\mathcal{H}, \mathcal{G})$ continues to mean bounded linear operators; an initialized plant G need not be linear as an input–output operator. For $K \in \mathcal{L}(\mathcal{H})$, define

$$(M_K w)(t) := Kw(t).$$

Feedback interconnections on Hilbert space

Definition 2 (Interconnection of G and Δ)**.** Given operators $G : L^n_{e,[a,b]} \to L^m_{e,[a,b]}$ and $\Delta : L^m_{e,[a,b]} \to L^n_{e,[a,b]}$, an admissible initial condition $x(0)$ for G, and signals $e \in L^n_{e,[a,b]}$ and $f \in L^m_{e,[a,b]}$, we say that $u \in L^n_{e,[a,b]}$ and $v \in L^m_{e,[a,b]}$ satisfy the interconnection defined by $[G, \Delta]$ if

$$v = Gu + f, \qquad u = \Delta v + e. \tag{1}$$

Here G is evaluated with the specified initial condition $x(0)$.

Definition 3 (Well-posedness)**.** Given G, Δ as above, we say the interconnection defined by $[G, \Delta]$ is well posed for all admissible plant initial conditions if, for every such $x(0)$ and any $e \in L^n_{e,[a,b]}$, $f \in L^m_{e,[a,b]}$, the following hold.

1. *Existence and uniqueness:* There exist unique $u \in L^n_{e,[a,b]}$, $v \in L^m_{e,[a,b]}$ satisfying (1) with that $x(0)$.
2. *Causality:* If u, v satisfy the interconnection for e, f, and $\hat{u}, \hat{v}$ satisfy it for $P_\tau e, P_\tau f$, with the same $x(0)$, then $P_\tau(u - \hat{u}) = 0$ and $P_\tau(v - \hat{v}) = 0$.

For a well-posed interconnection, retain the paper's notation

$$\begin{bmatrix} u \\ v \end{bmatrix} = F_{G,\Delta}\left(\begin{bmatrix} e \\ f \end{bmatrix}\right),$$

with the dependence on the specified $x(0)$ left implicit in G.

Definition 4 (Stability allowing for plant initial conditions)**.** We say the feedback system defined by $[G, \Delta]$ is stable allowing for plant initial conditions if it is well posed in the sense of Definition 3 and there exist constants $C, C_{\text{init}} \geq 0$ such that, for any $e \in L_Z^n$, $f \in L_Z^m$, and every admissible $x(0)$, the corresponding signals satisfy

$$\left\| \begin{bmatrix} u \\ v \end{bmatrix} \right\|_L \leq C \left\| \begin{bmatrix} e \\ f \end{bmatrix} \right\|_L + C_{\text{init}} \left\| x(0) \right\|_Z .$$

Both constants are independent of $e, f, x(0)$. Stability is robust over a class $\mathbf{\Delta}$ if well-posedness and this estimate hold for every $\Delta \in \mathbf{\Delta}$ with common constants.

This changes the paper's Definition 4 by retaining the possible output caused by the initial state. It is an input–output definition; it does not, by itself, assert convergence of hidden internal states.

Definition 5 (Hard IQC)**.** We say the operator $\Delta : L_{e,[a,b]}^m \to L_{e,[a,b]}^n$ satisfies the hard IQC defined by $\Psi : L_{e,[a,b]}^{m+n} \to L_{e,[a,b]}^{m+n}$ and $K \in \mathcal{L}(Z^{m+n})$ if, for any $T > 0$ and all $v \in L_{e,[a,b]}^m$,

$$\left\langle P_T \Psi \begin{bmatrix} v \\ \Delta v \end{bmatrix}, P_T M_K \Psi \begin{bmatrix} v \\ \Delta v \end{bmatrix} \right\rangle_L \geq 0. \tag{2}$$

The multiplier uses its specified zero initialization, and Δ uses the initialization specified by its input–output operator.

Definition 5 is unchanged apart from making initialization and the input dimension explicit. It does not require $x(0) = 0$ for the plant.

The main theorem

Theorem 6 (IQC theorem with nonzero plant initial conditions)**.** *If*

1. *$G : L_{e,[a,b]}^n \to L_{e,[a,b]}^m$ is causal and bounded on $L_{e,[a,b]}^n$ allowing for initial conditions, in the sense of Definition 1, with constants C_G, c_G;*
2. *$\Delta : L_{e,[a,b]}^m \to L_{e,[a,b]}^n$ and the interconnection defined by $[G, \Delta]$ is well posed as in Definition 3;*
3. *$\Psi : L_{e,[a,b]}^{n+m} \to L_{e,[a,b]}^{n+m}$ is causal and incrementally bounded, $\Psi(0) = 0$, and $K \in \mathcal{L}(Z^{n+m})$;*
4. *Δ satisfies the hard IQC defined by Ψ, K;*
5. *there exist $\varepsilon > 0$, $c_V \geq 0$, and a nonnegative function V satisfying $V(x(0)) \leq c_V \left\| x(0) \right\|_Z^2$ such that, for every admissible $x(0)$ and every $u \in L_{e,[a,b]}^n$,*

$$\begin{aligned} \left\langle P_T \Psi \begin{bmatrix} Gu \\ u \end{bmatrix}, P_T M_K \Psi \begin{bmatrix} Gu \\ u \end{bmatrix} \right\rangle_L \\ \leq -\varepsilon \left\| P_T u \right\|_L^2 + V(x(0)), \qquad T > 0. \end{aligned} \tag{3}$$

then the feedback system defined by $[G, \Delta]$ is stable in the sense of Definition 4. If these hypotheses hold with the same G bounds, Ψ, K, ε, c_V for every $\Delta \in \mathbf{\Delta}$, then the stability is robust over $\mathbf{\Delta}$ in the sense of Definition 4.

Proof. Define the shorthand notation $u_T := P_T u$ and $\langle u, v \rangle_T := \langle P_T u, P_T v \rangle_L$. Clearly, by Cauchy–Schwarz, $\langle u, v \rangle_T \leq \left\| u_T \right\|_L \left\| v_T \right\|_L$.

Now, because the interconnection defined by $[G, \Delta]$ is well posed, it suffices to establish the claimed bound uniformly over all finite horizons. For $e \in L_Z^n$, $f \in L_Z^m$, suppose that u, v satisfy the interconnection defined by G, Δ. Then

$$v = Gu + f, \qquad u = \Delta v + e.$$

By the hard IQC and (3),

$$\begin{aligned} \varepsilon \left\| u_T \right\|_L^2 \leq V(x(0)) + \left\langle \Psi \begin{bmatrix} v \\ \Delta v \end{bmatrix}, M_K \Psi \begin{bmatrix} v \\ \Delta v \end{bmatrix} \right\rangle_T \\ - \left\langle \Psi \begin{bmatrix} Gu \\ u \end{bmatrix}, M_K \Psi \begin{bmatrix} Gu \\ u \end{bmatrix} \right\rangle_T . \end{aligned}$$

Now, because Ψ and hence $M_K \Psi$ are incrementally bounded, with bounds C_Ψ and $C_{M_K \Psi} = C_\Psi \left\| K \right\|$, respectively, and $\Psi(0) = 0$, expanding the difference of the quadratic forms and applying Cauchy–Schwarz gives

$$\begin{aligned} \varepsilon \left\| u_T \right\|_L^2 \leq V(x(0)) + C_\Psi C_{M_K \Psi} \left(\left\| \begin{bmatrix} v - Gu \\ \Delta v - u \end{bmatrix}_T \right\|_L^2 \right. \\ \left. + 2 \left\| \begin{bmatrix} Gu \\ u \end{bmatrix}_T \right\|_L \left\| \begin{bmatrix} v - Gu \\ \Delta v - u \end{bmatrix}_T \right\|_L \right). \end{aligned}$$

Since $v - Gu = f$, $\Delta v - u = -e$, and

$$\left\| \begin{bmatrix} Gu \\ u \end{bmatrix}_T \right\|_L \leq (C_G + 1) \left\| u_T \right\|_L + c_G \left\| x(0) \right\|_Z ,$$

we conclude that

$$\begin{aligned} \varepsilon \left\| u_T \right\|_L^2 \leq V(x(0)) + C_\Psi C_{M_K \Psi} \left\| \begin{bmatrix} f \\ e \end{bmatrix}_T \right\|_L^2 \\ + 2 C_\Psi C_{M_K \Psi} (C_G + 1) \left\| u_T \right\|_L \left\| \begin{bmatrix} f \\ e \end{bmatrix}_T \right\|_L \\ + 2 C_\Psi C_{M_K \Psi} c_G \left\| x(0) \right\|_Z \left\| \begin{bmatrix} f \\ e \end{bmatrix}_T \right\|_L . \end{aligned}$$

Applying Young's inequality to the last two terms yields

$$\frac{\varepsilon}{2} \left\| u_T \right\|_L^2 \leq V(x(0)) + C_\Psi C_{M_K \Psi} c_G^2 \left\| x(0) \right\|_Z^2$$

$$+ 2C_\Psi C_{M_K\Psi} \left\| \begin{bmatrix} f \\ e \end{bmatrix}_T \right\|_L^2 + \frac{2(C_\Psi C_{M_K\Psi})^2 (C_G+1)^2}{\varepsilon} \left\| \begin{bmatrix} f \\ e \end{bmatrix}_T \right\|_L^2 .$$

Using $V(x(0)) \le c_V \|x(0)\|_Z^2$, define

$$\gamma_x^2 = \frac{2(c_V + C_\Psi C_{M_K\Psi} c_G^2)}{\varepsilon},$$
$$\gamma_d^2 = \frac{4C_\Psi C_{M_K\Psi}}{\varepsilon} + \frac{4(C_\Psi C_{M_K\Psi})^2 (C_G+1)^2}{\varepsilon^2}.$$

Thus

$$\|u_T\|_L \le \gamma_d \left\| \begin{bmatrix} f \\ e \end{bmatrix}_T \right\|_L + \gamma_x \|x(0)\|_Z .$$

Finally, since $v = Gu + f$,

$$\begin{aligned} \left\| \begin{bmatrix} u \\ v \end{bmatrix}_T \right\|_L &\le (C_G+1)\|u_T\|_L + c_G \|x(0)\|_Z + \|f_T\|_L \\ &\le \big((C_G+1)\gamma_d + 1\big) \left\| \begin{bmatrix} f \\ e \end{bmatrix}_T \right\|_L \\ &\quad + \big((C_G+1)\gamma_x + c_G\big) \|x(0)\|_Z . \end{aligned}$$

The constants are independent of T. Letting $T \to \infty$ proves the claim. No step sets $x(0) = 0$. □

Remark: The bound on G replaces the original zero-state boundedness assumption. For a linear plant it requires a uniformly finite-energy initial output in addition to the bounded zero-state input–output map. It does not follow from zero-state boundedness alone. The conclusion is initialized input–output stability, not an assertion of internal state convergence.

As in the paper, consider a PIE realization

$$\begin{aligned} \mathcal{T}\dot{x}(t) &= \mathcal{A}x(t) + \mathcal{B}u(t), \\ y(t) &= \mathcal{C}x(t) + \mathcal{D}u(t). \end{aligned} \tag{4}$$

The initial condition is now any admissible $x(0)$.

Lemma 7 (Sufficient version of KYP lemma with nonzero initial conditions)**.** *If*

1. $G : L^n_{e,[a,b]} \to L^m_{e,[a,b]}$ *is causal and bounded allowing for initial conditions in the sense of Definition 1;*
2. *there exist* $\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D} \in \Pi_4$ *such that, for any* $u \in L^n_{e,[a,b]}$, $y = Gu$ *implies that* y *satisfies* (4) *for a trajectory* x *with the specified admissible* $x(0)$*, and the integrated dissipation calculation below is valid for these trajectories;*
3. $K \in \Pi_4$ *and there exist* $\varepsilon > 0$ *and* $\mathcal{P} \in \Pi_4$ *such that* $\mathcal{P} \succeq 0$ *and*

$$\begin{bmatrix} \mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T} & \mathcal{T}^*\mathcal{P}\mathcal{B} \\ \mathcal{B}^*\mathcal{P}\mathcal{T} & \varepsilon I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} K \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0. \tag{5}$$

then, for any admissible $x(0)$ *and any* $u \in L^n_{e,[a,b]}$,

$$\begin{aligned} &\langle P_T G u, P_T M_K G u \rangle_L \\ &\quad \le -\varepsilon \|P_T u\|_L^2 + \langle \mathcal{T}x(0), \mathcal{P}\mathcal{T}x(0) \rangle_Z , \end{aligned}$$

for all $T > 0$.

Proof. Define $V(x) = \langle \mathcal{T}x, \mathcal{P}\mathcal{T}x \rangle$. Since $\mathcal{P} \succeq 0$ and $\mathcal{T}, \mathcal{P} \in \Pi_4$ are bounded, Cauchy–Schwarz and the operator norm give

$$0 \le V(x) = \langle x, \mathcal{T}^*\mathcal{P}\mathcal{T}x \rangle \le \|\mathcal{T}^*\mathcal{P}\mathcal{T}\| \, \|x\|_Z^2 .$$

Suppose that $u \in L^n_{e,[a,b]}$ and $y(t) = (Gu)(t)$ for a trajectory x satisfying (4) with arbitrary admissible $x(0)$. By inequality (5), multiplying on both sides by $\begin{bmatrix} x(t) \\ u(t) \end{bmatrix}$ in the quadratic form gives

$$\begin{aligned} &\langle \mathcal{T}\dot{x}(t), \mathcal{P}\mathcal{T}x(t) \rangle + \langle \mathcal{T}x(t), \mathcal{P}\mathcal{T}\dot{x}(t) \rangle \\ &\quad + \varepsilon \|u(t)\|_Z^2 + \langle y(t), Ky(t) \rangle_Z \\ &= \dot{V}(x(t)) + \varepsilon \|u(t)\|_Z^2 + \langle y(t), Ky(t) \rangle_Z \le 0. \end{aligned}$$

Integrating in time, we obtain

$$\begin{aligned} \varepsilon \|P_T u\|_L^2 + \int_0^T \langle y(t), Ky(t) \rangle_Z \, dt &\le -V(x(T)) + V(x(0)) \\ &\le V(x(0)), \end{aligned}$$

since $V(x(T)) \ge 0$. We conclude that

$$\begin{aligned} &\langle P_T G u, P_T M_K G u \rangle_L \\ &= \int_0^T \langle y(t), Ky(t) \rangle_Z \, dt \\ &\le -\varepsilon \|P_T u\|_L^2 + V(x(0)) \\ &= -\varepsilon \|P_T u\|_L^2 + \langle \mathcal{T}x(0), \mathcal{P}\mathcal{T}x(0) \rangle_Z . \end{aligned}$$

The initial term is retained throughout. □

Remarks: The cross blocks in (5) include $\mathcal{T}^*\mathcal{P}\mathcal{B}$ and $\mathcal{B}^*\mathcal{P}\mathcal{T}$, as required by the storage $V(x) = \langle \mathcal{T}x, \mathcal{P}\mathcal{T}x \rangle$ and the stated dynamics. The supplied PDF prints $\mathcal{P}\mathcal{B}$ and $\mathcal{B}^*\mathcal{P}$ instead. For mild solutions, the integrated inequality must be justified by regularity or extension from regular solutions.

Applying this lemma to the augmented plant–multiplier realization, as in Corollary 8 of the original paper, supplies (3), with the initial-storage allowance evaluated at $\begin{bmatrix} x(0) \\ 0 \end{bmatrix}$. The initialized bound on the nominal G is still a separate hypothesis; the lemma does not establish it.

One way to compute an admissible c_V is to impose

$$0 \preceq \mathcal{T}^*\mathcal{P}\mathcal{T} \preceq c_V I$$

together with (5). Since $\mathcal{T}$ is fixed, the upper bound is linear in $\mathcal{P}, c_V$. Alternatively take $c_V = \|\mathcal{T}\|^2 \|\mathcal{P}\|$. No coercive lower bound on the storage is needed for the initialized input–output conclusion of Theorem 6.

Examples verifying the conditions

Example 1 (Verifying initialized boundedness by a gain LPI)**.** For a PIE realization (4), suppose one can find $\mathcal{R} \in \Pi_4$, $\mathcal{R} \succeq 0$, and constants $\gamma, r_0 \geq 0$ satisfying

$$\begin{bmatrix} \mathcal{T}^*\mathcal{R}\mathcal{A} + \mathcal{A}^*\mathcal{R}\mathcal{T} & \mathcal{T}^*\mathcal{R}\mathcal{B} \\ \mathcal{B}^*\mathcal{R}\mathcal{T} & -\gamma^2 I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0,$$
$$0 \preceq \mathcal{T}^*\mathcal{R}\mathcal{T} \preceq r_0 I.$$

Multiplication by $\begin{bmatrix} x(t) \\ u(t) \end{bmatrix}$ and integration, justified for the chosen realization, give

$$\|P_T G u\|_L^2 \leq \gamma^2 \|P_T u\|_L^2 + r_0 \|x(0)\|_Z^2 .$$

Consequently Definition 1(3) holds with $C_G = \gamma$, $c_G = \sqrt{r_0}$. These are separate sufficient gain conditions for Theorem 6's first assumption. The operator $\mathcal{R}$ need not equal the IQC storage $\mathcal{P}$. For fixed γ, these are LPIs in $\mathcal{R}, r_0$; one can also search over γ^2 as a scalar variable.

Example 2 (The initialized bound for a stable nominal realization)**.** Consider a linear realization on a Hilbert state space, with bounded input and output operators B, C, D. Suppose its state-transition semigroup satisfies

$$\|S(t)\| \leq M e^{-\alpha t}, \qquad M, \alpha > 0.$$

For any admissible initial state, its output is

$$(Gu)(t) = CS(t)x(0) + Du(t) + \int_0^t CS(t-s)Bu(s)\, ds.$$

The initial-output term satisfies

$$\|CS(\cdot)x(0)\|_L \leq \frac{M\|C\|}{\sqrt{2\alpha}} \|x(0)\|_Z .$$

The convolution kernel has L_1 norm at most $M\|C\|\|B\|/\alpha$. Young's convolution inequality, applied to the truncated input, therefore verifies Definition 1(3) with

$$C_G = \|D\| + \frac{M\|C\|\|B\|}{\alpha},$$
$$c_G = \frac{M\|C\|}{\sqrt{2\alpha}}.$$

This is a sufficient way to verify the added assumption, not an assertion that exponential nominal stability is necessary. If the realization has an unbounded input or observation operator, appropriate admissibility estimates must replace these bounded-operator formulas.

Example 3 (A complete scalar certificate with arbitrary initial state)**.** Consider

$$\dot{x}(t) = -2x(t) + u(t), \qquad y(t) = x(t), \qquad x(0) \in \mathbb{R}.$$

Example 2 gives $C_G = 1/2$ and $c_G = 1/2$. Let $\mathbf{\Delta}$ consist of the memoryless operators $(\Delta v)(t) = \phi(v(t))$, where $\phi(0) = 0$ and ϕ is globally Lipschitz with constant at most one. Then $|\phi(v)| \leq |v|$, so every member satisfies the same hard IQC with

$$\Psi = I, \qquad K = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Indeed,

$$\int_0^T \big(|v(t)|^2 - |\phi(v(t))|^2\big) dt \geq 0.$$

Here Ψ is causal, $\Psi(0) = 0$, and its incremental bound is one.

For arbitrary $e, f \in L_{2e}$, the feedback equations give

$$\dot{x}(t) = -2x(t) + \phi(x(t) + f(t)) + e(t).$$

Its right-hand side is globally Lipschitz in x, uniformly in time, and has locally integrable forcing. Consequently it has a unique global absolutely continuous solution for every $x(0)$; causality follows from uniqueness. This verifies Definition 3 without requiring an algebraic-loop assumption, because the plant has no feedthrough.

Apply Lemma 7 to the graph output $\begin{bmatrix} y \\ u \end{bmatrix}$, as in the paper's augmented construction. Its output operators are $\mathcal{C} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathcal{D} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, while $\mathcal{T} = 1$, $\mathcal{A} = -2$, and $\mathcal{B} = 1$. Choose

$$\mathcal{P} = 2, \qquad \varepsilon = \tfrac{1}{4}.$$

The left-hand side of (5) becomes

$$\begin{bmatrix} -8 & 2 \\ 2 & 1/4 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -7 & 2 \\ 2 & -3/4 \end{bmatrix} \prec 0.$$

The first diagonal entry is negative and the determinant is $21/4 - 4 = 5/4 > 0$, verifying strict negative definiteness. The storage is

$$V(x) = 2x^2, \qquad c_V = 2.$$

Lemma 7 consequently supplies precisely the initialized separation condition required in Theorem 6:

$$\int_0^T \big(|(Gu)(t)|^2 - |u(t)|^2\big) dt \leq -\tfrac{1}{4} \int_0^T |u(t)|^2 dt + 2|x(0)|^2.$$

All constants are common to the uncertainty class. Theorem 6 therefore gives robust input–output stability allowing for every $x(0)$. The time-domain calculations in Lemma 7 are valid almost everywhere for these absolutely continuous trajectories and can be integrated.

Example 4 (An infinite-dimensional initial-output bound)**.** On the Hilbert space $L_2(0,1)$ consider the heat equation

$$x_t(s,t) = x_{ss}(s,t) + u(s,t),$$
$$x(0,t) = x(1,t) = 0, \qquad y(s,t) = x(s,t).$$

The initial profile $x(\cdot,0)$ may be any element of $L_2(0,1)$. The Dirichlet heat semigroup has eigenvalues $-j^2\pi^2$, $j = 1,2,\ldots$, and satisfies $\|S(t)\| \le e^{-\pi^2 t}$. Taking $B = C = I$, $D = 0$ in Example 2 gives

$$C_G = \frac{1}{\pi^2}, \qquad c_G = \frac{1}{\sqrt{2}\,\pi}.$$

Thus the initialized boundedness assumption is meaningful for arbitrary nonzero infinite-dimensional initial profiles, not just for ODE states.

For a pointwise nonlinearity ϕ as in Example 3, the induced operator on $L_2(0,1)$ is globally Lipschitz with constant one. The resulting semilinear heat equation with $e, f \in L_{2e}$ has unique global mild solutions. This verifies the corresponding well-posedness requirement. For regular initial profiles and inputs, energy differentiation is available directly. To use a PIE storage estimate for all mild solutions, first prove it for regular data and pass to the limit using density, continuous dependence, and boundedness of the storage operators. A particular PIE realization and its LPI must still be verified; the semigroup bound alone does not verify (5). The initial-state norm and admissible profiles must match that PIE realization: a general physical L_2 profile is not automatically an admissible initial fundamental state if that state contains derivatives.

Example 5 (A dynamic multiplier satisfying the IQC conditions)**.** Let $(\Delta v)(t) = d\,v(t)$ for an unknown constant $d \in [-1,1]$. Choose the zero-initialized stable filter H with transfer function $H(s) = 1/(s+1)$, acting componentwise if the signals are Hilbert-valued, and take

$$\Psi = \begin{bmatrix} H & 0 \\ 0 & H \end{bmatrix}, \qquad K = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}.$$

The filter is causal, bounded, incrementally bounded, and maps zero to zero. Its L_2 gain is at most one. Since d is constant and H is linear, $H(dv) = dHv$. Therefore

$$\left\langle P_T\Psi \begin{bmatrix} v \\ \Delta v \end{bmatrix}, P_T M_K \Psi \begin{bmatrix} v \\ \Delta v \end{bmatrix} \right\rangle_L = (1-d^2)\int_0^T \|(Hv)(t)\|_Z^2\,dt \ge 0.$$

This verifies the hard IQC with the same multiplier for every allowed d. The argument uses a constant scalar uncertainty; it does not establish the same filtered IQC for a general time-varying $d(t)$. The plant still needs the initialized bound, well-posedness, and the separation certificate in Theorem 6.

References

[1] A. Talitckii, M. M. Peet, and P. Seiler, *Integral Quadratic Constraints with Infinite-Dimensional Channels*, 2023 American Control Conference, pp. 1576–1583, 2023, doi: 10.23919/ACC55779.2023.10156335.